Protecting the FPGA IPs against Higher-order Side Channel Attacks using Dynamic Partial Reconfiguration

Ario Kian
Faculty of Computer Sci. and Eng.
Shahid Beheshti University
Tehran 1983963113, IRAN
a.kianazad@mail.sbu.ac.ir

Hamed Hosseintalaee
Faculty of Computer Sci. and Eng.
Shahid Beheshti University
Tehran 1983963113, IRAN
h_hosseintalaee@sbu.ac.ir

Ali Jahanian
Faculty of Computer Sci. and Eng.
Shahid Beheshti University
Tehran 1983963113, IRAN
jahanian@sbu.ac.ir

***Abstract*—Power Attacks have become a serious concern for secure FPGA-based embedded system design. However, considerable countermeasures have been introduced to improve the resistance of FPGA devices against Differential Power Analysis (DPA) and Correlation Power Analysis (CPA) attacks during recent years. In this work, we proposed a novel countermeasure which utilizes the Dynamic Partial Reconfiguration (DPR) property of the FPGA devices to obfuscate the leaked information. DPR capability allows updating a certain part of the running design on the device while other parts remain functioning without interruption. We proposed a new masking scheme using this technology in which masking function is changed dynamically at runtime to violate the assumptions needed for a successful power analysis attack. The proposed method not only overcomes to the first-order DPA attacks but also increases the resistance against the second-order DPA attacks. Our experimental results show that the proposed method has considerable security improvement against first and second order SCA attacks with negligible overheads. Moreover it can be complemented with any other desired countermeasures as well.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

Studies have shown that the security of the cryptographic algorithm itself does not guarantee the security of the secret information since there are other threats originating from the implementation of the algorithm known as side-channel analysis (SCA) attacks.

These kind of attacks exploit physical information leakages such as power consumption, electromagnetic radiations and timing characteristics of the implemented design. Considering the strong statistical backgrounds shown in [1] and [2], alongside the simplicity of the attack strategies and ability of physical access to these systems, power analysis attacks as a kind of SCA attacks have been very practical and a serious concern for Field Programmable Gate Array (FPGA) developers.

Considerable countermeasures have been introduced against SCAs since these kinds of attacks were first introduced in [3].

978-1-7281-8111-0/20/$31.00 ©2020 IEEE

Dynamic Partial Reconfiguration (DPR) is one of the features of modern FPGA devices that have been used in several studies to develop countermeasures against power analysis attacks. To the best of our knowledge, the first work that considered to use this feature was [4] introducing temporal Jitter by randomly relocating the sub-operations of the block cipher through the FPGA fabric. But according to [5], this timing disarrangement can be overcome by the attacker using advanced alignment filtering functions.

In [6], the authors are using Configurable Look-Up Tables (CFGLUT) in Spartan-6 FPGAs as a dynamic mechanism to change logic randomly. The main idea is to split SBox components into two new SBoxes with random mappings in order to store some random unknown values in registers that is produced by the first random mapping while the second mapping computes the actual value of the original SBox. Using CFGLUTs increases the hardware resource usage of FPGA fabric rather than the actual partial reconfiguration of the device using partial bitstreams. Consequently, for block ciphers which use large SBox components such as AES, this mechanism is not resource-efficient. The countermeasure based on the dynamic random SBox mappings is not very effective due to the existence of information leakage at combinational logic and none-registered values as well.

Authors of [7], used a ring oscillator based power estimator alongside the block cipher. The proposed system changes the size and the number of hardware resources dedicated to a Gaussian noise generator based on the estimated power consumption using partial reconfiguration that changes the amplitude and variance of the generated noise dynamically. This paper does not present the proof of its method. Authors of [8] created a set of different instances of each functional component of the block cipher which produces different power traces due to the difference in the implementation characteristics. These different instances are swapped in every execution of the block cipher using partial reconfiguration. But similar to [7], they have not shown any proof about their work.

The presented contribution in [9] is very similar to [8] but they kept using the same RTL and to achieve the physical diversity

Authorized licensed use limited to: Universidade Tecnologica Federal do Parana. Downloaded on October 14,2026 at 20:06:47 UTC from IEEE Xplore. Restrictions apply.

between the configurations, they used different placement and routing on the FPGA fabric. According to their report, this method remarkably improves the resistance against SCA attacks but other parts of the implemented algorithm in the first round such as the first byte in the ShiftRow component may be vulnerable as the implementation remains unchanged. In the previously discussed researches, there are some limits such as advance of attacking tools and techniques, timing misalignment in the affection of the countermeasure, the existence of leakage at combinational logic, and not covering all the possible points targeted by an attacker.

In this work, we present a novel masking approach that uses the DPR capability to change the masking functions used in the masking process randomly in order to decrease the maximum correlation coefficient needed for a successful second-order DPA attack introduced in [10] as well as reasonable protection against first-order Differential Power Analysis (DPA) and Correlation Power Analysis (CPA) attacks. We applied our approach to a conventional masking scheme and implemented it on AES block cipher using a Zynq-7020 MPSoC device. The proposed approach is evaluated by attacking the protected system presenting experimental and statistical results.

The remainder of this article is organized as follows. Section II explains general concepts and considerations about developing countermeasures against SCA attacks. Our general approach and fundamental requirements are presented precisely in Section III. The evaluation of the suggested method is demonstrated in section IV and finally Section V concludes the paper.

II. Countermeasures against SCA

DPA and CPA are the most popular types of Power analysis attacks. According to [1], it is usually sufficient to know the cryptographic algorithm implemented on the device and the secret key can be revealed even if the recorded power traces are extremely noisy.

The main goal of every countermeasure is to reduce the statistical dependencies between the processed intermediate value and the power consumed by the device corresponding to that value. Existing countermeasures are categorized as Hiding and Masking. The main idea of hiding is to change the power consumption characteristics of the device in order to bury the exploitable part of the power traces for the attacker with a lot of non-relevant information such as noise and timing disarrangements. While in hiding techniques, the cryptography subsystem calculates the same intermediate values as the referenced algorithm, masking techniques try to randomize the intermediate values in a way that the effort to make the correct key guess is maximized. More precisely, in masking techniques, all the intermediate values V that are generated during the encryption/decryption process are randomized to a masked value V_m using a randomly generated value m and a masking function * as shown in (1).

$$V_m = V * m \tag{1}$$

The masking function $*$ used in (1) is normally the Boolean XOR operation in symmetric cryptographic algorithms such as AES. In case of AES, SBox components in the SubByte operations need to be changed since they are non-Linear functions in order to produce the same ciphertext as the non-masked version. Consequently, designer must create SBox Components S' in which $S'(x \oplus m) = S(x) \oplus m$ as suggested in early AES masking studies. By following the changes applied to the randomly generated mask value M, we can simply calculate the last mask value M^n used in the last computed state of the AES. To obtain the original expected ciphertext, it is sufficient to exclusive-or the final produced value with the last calculated mask value M^n. Masking methods provide a proper resistance against first-order attacks, but they are vulnerable to second and higher-order attacks. Generally, in second-order attacks, the attacker will choose two intermediate values corresponding to two sample points in power traces, in a way that the effect of the masking process is bypassed. In the following section, we have described a practical second order DPA attack introduced by [10]. Although these attacks can be negated by applying several random masks to each intermediate values, but this process drastically decreases the performance of the system and increase the resource usage on the FPGA fabric.

III. The Proposed Method

Authors of [10] introduce a practical second-order DPA attack which relies on the assumption that the power consumption C of the device when data a is processed, is proportional to the Hamming weight of a.

$$C(a) \approx HW(a) \tag{2}$$

Equation 3 for data values $a, b \in \{0,1\}$ holds with probability of one. In case of $a, b \in \{0,1\}^n$, $n > 1$, the relation (3) only holds for some possible values. But increasing the number of samples makes the existing errors negligible.

$$HW(a \oplus b) = |HW(a) - HW(b)| \tag{3}$$

Consequently, the value $|C(a) - C(b)|$ can be estimated with $HW(a \oplus b)$, if the number of given samples is sufficient. In this situation if the attacker chooses two suitable intermediate values such as input and output of a masked SBox, the effect of the random mask value would be bypassed as shown in (5):

$$\begin{aligned} a &= p_i \oplus k_i \oplus m_i \\ b &= S(p_i \oplus k_i) \oplus m_i \end{aligned} \tag{4}$$

where

$$\begin{aligned} &|C(p_i \oplus k_i \oplus m_i) - C(S(p_i \oplus k_i) m_i)| \\ &\approx HW(p_i \oplus k_i \oplus m_i \oplus S(p_i \oplus k_i) \oplus m_i) \\ &= HW(p_i \oplus k_i \oplus S(p_i \oplus k_i)) \end{aligned} \tag{5}$$

It can be concluded that the consistency of using XOR as a Boolean masking function through the whole masking process is the reason for the practicality of this attack.

In the proposed method in this paper, we use dynamic partial

Authorized licensed use limited to: Universidade Tecnologica Federal do Parana. Downloaded on October 14,2026 at 20:06:47 UTC from IEEE Xplore. Restrictions apply.

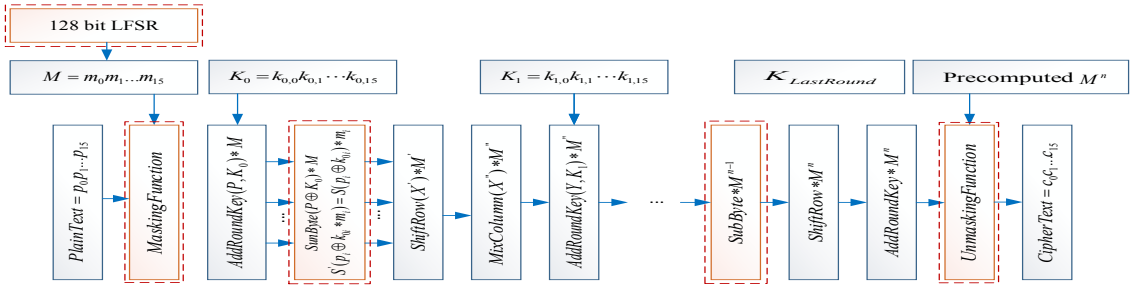


Fig. 1. Dynamic Masking Scheme

reconfiguration to replace the current masking functions with a randomly selected associative and self-inverse function * which must satisfy the property given in (6).

$$HW(x * m \oplus y * m) = HW(x * y * 0) \neq HW(x \oplus y) \tag{6}$$

In this situation, the masking scheme is changed as Fig. 1. In this figure, dashed blocks are reconfigurable and others are implemented statically. According to Fig. 1, a set of proper functions (e.g. *) are needed to be defined as masking functions. Also for each function $*$ and random value m_i generated by a pseudo-random number generator (LFSR), a new set of SBox components S' are pre-computed and implemented according to (7) to be replaced with by the current SBox components in every execution of the algorithm. S' should meet the following relation.

$$S'(p_i \oplus k_i * m_i) = S(p_i \oplus k_i) * m_i \tag{7}$$

We implemented these blocks as reconfigurable modules to update the masking functions and corresponding SBox components. For every new execution of the AES, one of the pre-defined functions is selected randomly. According to the randomly generated value m_i and the selected function, the partial bitstreams for blocks *maskingFunction*, *unmaskingFunction* and S' are loaded into the device.

IV. Evaluation and Discussion

We have implemented our countermeasure based on a conventional scenario that uses two functions XOR and XNOR as the randomly switching masking functions. To satisty the condition discussed in (6), the masked SBox components S' of each round of the algorithm, must be generated based on the XNOR function if the masking function effected at the input of the SBox components in the corresponding round is the XOR, and vice versa as shown in Fig. 2. Considering this constraint, the power consumption would be estimated by the attacker based on (8) and (9).

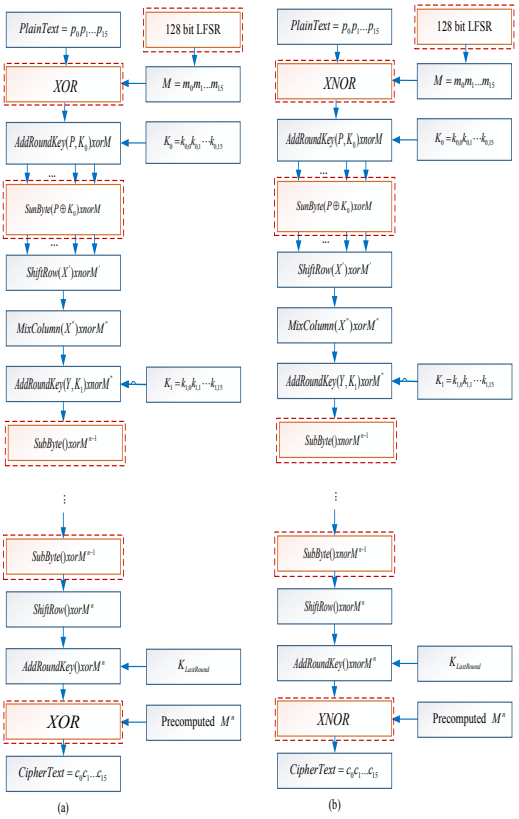


Fig. 2. Dynamic Masking Scheme implemented by XOR and XNOR functions

$$\begin{aligned} HW(S(p_i \oplus k_i) \odot m_i \oplus p_i \oplus k_i \oplus m_i) \\ = HW(S(p_i \oplus k_i) \odot p_i \oplus k_i \oplus 0) \\ = HW(S(p_i \oplus k_i) \odot p_i \oplus k_i) \\ \neq HW(S(p_i \oplus k_i) \oplus p_i \oplus k_i) \end{aligned} \tag{8}$$

and

$$\begin{aligned} HW(S(p_i \oplus k_i) \oplus m_i \oplus p_i \oplus k_i \odot m_i) \\ = HW(S(p_i \oplus k_i) \oplus p_i \oplus k_i \odot 0) \\ = HW(S(p_i \oplus k_i) \oplus \overline{(p_i \oplus k_i)}) \\ \neq HW(S(p_i \oplus k_i) \oplus p_i \oplus k_i) \end{aligned} \tag{9}$$

Authorized licensed use limited to: Universidade Tecnologica Federal do Parana. Downloaded on October 14,2026 at 20:06:47 UTC from IEEE Xplore. Restrictions apply.

According to (8) and (9), the assumptions that are needed for a successful second-order DPA is violated and the attacker would not be able to compute the correct modeled power consumption values in order to achieve a proper correlation coefficient with the correct key hypothesis.

The Z-Turn development board is used for the implementation and evaluation of the protected method in this paper. Power consumption traces are extracted over a 1Ω resistor at the GND line of the device using the GDS-2304A oscilloscope with 2Gs/S and 2MPoints memory. We performed a first-order DPA/CPA attack on the first SBox output in the first round of the AES Encryption Algorithm. In Non-protected implementation, we had a successful CPA attack by analyzing 1 million power traces, while by recording over 38 million power traces, we could not get any significant correlation needed to detect the correct key hypothesis in case of protected design. To evaluate the resistance against second-order DPA/CPA the minimum number of traces for both fixed XOR masking function and our random masking function are compared. The minimum number of traces can be estimated by (10), presented by [1] and [10].

$$N = 3 + 8(\frac{Z_{1-\alpha}}{ln\frac{1+\rho}{1-\rho}}) \tag{10}$$

In (10), $Z_{1-\alpha}$ is the probability of achieving a successful attack which is usually set to $Z_{0.9999} = 3.719$ and ρ is the correlation coefficient between the correct key hypothesis and the power traces. To estimate the correlation coefficient ρ used in (10), the ideal model is assumed where $C(a) = HW(a)$ as suggested to do in [10]. Equations (11) and (12) show the estimation of the correlations for fixed masking function and dynamic random masking function respectively.

$$\rho_{fixed} = Corr(HW(S(p \oplus k) \oplus p \oplus k), \\ |HW(S(p \oplus k) \oplus m) - HW(p \oplus k \oplus m)|) \tag{11}$$

and

$$\rho_{dynamic} = Corr(HW(S(p \oplus k) \oplus p \oplus k), \\ |HW(S(p \oplus k) * m) - HW(p \oplus k * m)|) \tag{12}$$

The estimated ρ for fixed XOR masking function and five different executions of dynamic masking function ($\rho_{dynamic(exc1)}, ..., \rho_{dynamic(exc5)}$) and the corresponding minimum number of traces are given in Table I.

According to results shown in Table I, using only two functions that are swapping randomly and a slight change in the masking scheme, increases the number of needed traces significantly that proves that the proposed method improves the security of design against SCA.

TABLE I
COMPUTED CORRELATION COEFFICIENTS IN IDEAL MODEL FOR DIFFERENT RUNS OF THE SYSTEM

ρ	**Calculated Correlation**	**Number Of needed traces**
ρ_{fixed}	0.201	661
$\rho_{dynamic}$ (exc1)	0.032	30738
$\rho_{dynamic}$ (exc2)	0.069	5647
$\rho_{dynamic}$ (exc3)	0.057	9147
$\rho_{dynamic}$ (exc4)	0.066	6550
$\rho_{dynamic}$ (exc5)	0.063	6680

V. CONCLUSION

In this work, we proposed a novel countermeasure which makes use of the dynamic partial reconfiguration of modern FPGAs to change the Boolean masking function of a masked AES algorithm randomly in order to increase the resistance of the implemented system against both first-order and second-order DPA/CPA attacks. The main idea has been tested by implementing a simple scenario in which the masking function is randomly switched between XOR and XNOR function with some additional constraints. Results of practical and statistical evaluations have shown a remarkable improvement in resistance against both first-order and second-order DPA attacks. Besides the considerable improvement our method brings to the security of the system, it can be complemented with any other desired countermeasures to achieve better resistance against higher order DPA/CPA attacks and any other SCA attacks as well.

REFERENCES

[1] S. Mangard, E. Oswald and T. Popp, Power Analysis Attacks, in Revealing the Secrets of Smart Cards. Springer, 2007, p. 351.

[2] P. Kocher, J. Jaffe, B. Jun and P. Rohatgi, Introduction to differential power analysis. Springer, p. 23, J Cryptogr Eng (2011) 1: 5. https://doi.org/10.1007/s13389-011-0006-y.

[3] P. Kocher, J. Jaffe, B.Jun, Differential Power Analysis, In: Wiener M. (eds) Advances in Cryptology - CRYPTO' 99. CRYPTO 1999. Lecture Notes in Computer Science, vol 1666. Springer, Berlin, Heidelberg.

[4] N. Mentens, B. Gierlichs, I. Verbauwhede, Power and Fault Analysis Resistance in Hardware through Dynamic Reconfiguration. In: Oswald E., Rohatgi P. (eds) Cryptographic Hardware and Embedded Systems - CHES 2008. CHES 2008. Lecture Notes in Computer Science, vol 5154. Springer, Berlin, Heidelberg.

[5] T. Güneysu, A. Moradi, Generic Side-Channel Countermeasures for Reconfigurable Devices. In: Preneel B., Takagi T. (eds) Cryptographic Hardware and Embedded Systems - CHES 2011. CHES 2011. Lecture Notes in Computer Science, vol 6917. Springer, Berlin, Heidelberg.

[6] P. Sasdrich, A. Moradi, O. Mischke and T. Guneysu, Achieving Side-Channel Protection with Dynamic Logic Reconfiguration on Modern FPGAs. in 2015 IEEE International Symposium on Hardware Oriented Security and Trust (HOST), Washington, DC, USA, 2015.

[7] G. Dessouky and A.-R. Sadeghi, POSTER: Exploiting Dynamic Partial Reconfiguration for Improved Resistance Against Power Analysis Attacks on FPGAs. in Proceedings of the 9th ACM Conference on Security & Privacy in Wireless and Mobile Networks, Darmstadt, Germany, July 18 - 20, 2016.

[8] N. Bete, F. Saqib, C. Patel, R. Robucci and J. Plusquellic, Side-channel Power Resistance for Encryption Algorithms using Dynamic Partial Reconfiguration (SPREAD). in HOST Symposium, 2017.

[9] B. Hettwer, J. Petersen, S. Gehrer, H. Neumann and T. Güneysu, Securing Cryptographic Circuits by Exploiting Implementation Diversity and Partial Reconfiguration on FPGAs. in Design, Automation & Test in Europe Conference & Exhibition (DATE), Florence, Italy, 25-29 March 2019.

[10] E. Oswald, S. Mangard, C. Herbst, S. Tillich, Practical Second-Order DPA Attacks for Masked Smart Card Implementations of Block Ciphers. In: Pointcheval D. (eds) Topics in Cryptology - CT-RSA 2006. CT-RSA 2006. Lecture Notes in Computer Science, vol 3860. Springer, Berlin, Heidelberg.

Authorized licensed use limited to: Universidade Tecnologica Federal do Parana. Downloaded on October 14,2026 at 20:06:47 UTC from IEEE Xplore. Restrictions apply.